

ResolveIQ

AI-Powered Frictionless Dispute
& Chargeback Resolution

Round 1 Submission

Challenge Track: Frictionless Dispute & Chargeback Resolution

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< 5 min

Resolution time

70%

Cases auto-resolved

\$40+

Saved per dispute

2×

NPS uplift

The Dispute Experience is Broken

01

14–45 Days

Average resolution
cycle today

02

Manual & Opaque

No visibility, no context,
no explanation

03

\$25–\$50

Cost per manually
handled dispute

Card Members

Drop disputes due to friction

Merchants

Blanket evidence requests

Issuers

Agent bandwidth wasted

Regulators

Hard to audit at scale

Introducing ResolveIQ

Three intelligent layers — one seamless resolution in under 5 minutes

01

Evidence Autopilot

Auto-collects transaction metadata,
merchant data, emails & receipts via NLP
— baseline is never zero



02

FairScore™ Model

Dual-perspective AI weighs evidence
from both sides across 6 dimensions —
not a black-box classifier



03

Explainability Engine

LLM generates plain-English decision
reasoning grounded in FairScore™
output — for both parties

Evidence Autopilot

Zero-Touch Evidence Collection

Transaction Metadata

Timestamp · MCC · Amount ·
Terminal ID

Merchant Data

Delivery confirmations · Receipts
· Policy

Email NLP Parser

spaCy + DistilBERT entity
extraction

Chargeback Codes

Auto-mapped AMEX reason
code(s)

Case Similarity

RAG lookup of past resolved
disputes

Member History

Relationship tenure · Dispute
frequency

Key Insight

Most dispute tools start from an empty form.

Evidence Autopilot means the baseline is
NEVER zero — dramatically reducing
abandonment and incomplete submissions.

Both parties can supplement, but never need
to start from scratch.

FairScore™ Weighing Model

Dual-perspective AI — explicit scoring for both card member AND merchant

Evidence Strength

Quality of docs from each side

Reason Code Fit

Claim vs. chargeback category

Merchant History

Win/loss rate · flags abusers

Member History

Dispute frequency check

Case Precedent

RAG vector similarity to past cases

Policy Compliance

Windows · amount thresholds

Output

0–100

Confidence Score

**XGBoost
+ SHAP**

Transparent model
(not a black box)

→ Favour Card Member

→ Favour Merchant

→ Escalate to Human

Explainability Engine

"Why was this decided?" — every outcome explained in plain English, to both parties

Card Member View

We resolved this dispute in your favour because:

- The merchant couldn't provide a signed delivery confirmation within the required window
- Transaction amount didn't match your receipt
- 94% of similar cases resolved in card member's favour

Decision per AMEX Chargeback Code 4513.

Merchant View

We were unable to resolve this in your favour because:

- Delivery confirmation post-dates the dispute filing
- Item description on receipt doesn't match the transaction

We recommend reviewing your order fulfilment timestamps and aligning receipt line items with transaction records.



40% fewer appeals



Regulatory-ready (CFPB, EU AI Act)



Grounded in FairScore™ weights —
not post-hoc

What Makes ResolveIQ Different

01 Zero-Touch Evidence Collection

Evidence is auto-pulled — no manual upload required as baseline. Biggest friction reducer in the dispute funnel.

02 Dual-Perspective Fairness Model

Explicitly models both sides. Most tools default to protecting card members. We model merchant fairness too.

03 Explainable AI by Design

Reasoning is structural, not bolted on. Maps directly to EU AI Act Art. 22 & CFPB explainability requirements.

04 RAG-Based Precedent Engine

Past resolved disputes as a knowledge base — decisions are consistent, improvable, and precedent-cited.

05 Immutable Audit Trail

Append-only, SHA-256 hash-chained log. Every decision, evidence item, and action is verifiably recorded.

Technical Architecture



Layer	Technology
Frontend	React · Tailwind CSS · WebSockets
Backend	Python FastAPI · Node.js
NLP / AI	spaCy · Hugging Face DistilBERT
Decision Model	XGBoost + SHAP values
Precedent	pgvector on PostgreSQL (RAG)
LLM Layer	Claude API / GPT-4
Cloud	AWS Lambda · S3 · RDS · SNS
Audit Store	Append-only PostgreSQL + SHA-256



**AWS
Cloud**

Lambda · S3
RDS · SNS · SES

Three Portals. One Seamless Experience.

Card Member Portal

- ✓ One-click dispute filing
- ✓ Real-time status tracker (WebSocket)
- ✓ Plain-language decision view
- ✓ Evidence supplement upload

Merchant Portal

- ✓ Automated dispute notification
- ✓ Targeted evidence request (not blanket)
- ✓ Deadline countdown + submission
- ✓ Decision rationale + appeal option

AMEX Ops Dashboard

- ✓ Human review queue (low-confidence)
- ✓ Bulk audit log + export
- ✓ FairScore™ model monitoring
- ✓ Fairness metrics + drift alerts

Implementation Roadmap

Modular, feasible, production-ready in 10 weeks

Phase 1

Weeks 1–3

Foundation

- Core dispute filing flow
- Transaction evidence collection
- FairScore™ v1 (rule-based)
- Card member & merchant portals MVP

Phase 2

Weeks 4–6

AI Layer

- NLP evidence parser (spaCy)
- XGBoost FairScore™ model
- pgvector precedent engine
- Synthetic training data pipeline

Phase 3

Weeks 7–8

Explainability

- LLM decision explanations
- SHAP integration
- Audit log with hash chaining
- Appeal flow

Phase 4

Weeks 9–10

Ops & Polish

- AMEX Ops dashboard
- Model monitoring (Prometheus)
- Fairness metrics
- End-to-end testing

Off-the-shelf AI: No research-level ML needed

Modular: Each layer independently deployable

Human-in-loop: Low-confidence → human review

Measurable Value for AMEX

95%

Reduction in
resolution time

\$28M+

Annual ops savings
(1M disputes)

40%

Fewer
appeals

2×

Merchant
satisfaction

Cost Reduction

70% auto-resolution rate → \$28M+ annual savings on 1M dispute volume

Card Member Retention

Dispute experience is a top driver of churn — fast resolution improves loyalty

Merchant Relations

Fair, transparent process strengthens AMEX's merchant network vs competitors

Regulatory Compliance

Explainable AI + immutable audit trail future-proofs against CFPB & EU AI Act

ResolveIQ

"Dispute resolution is not a back-office problem — it is a front-line customer experience and a direct driver of card member and merchant loyalty."

01

Fast

14 days → < 5 minutes
for 70% of cases

02

Fair

Dual-perspective model
explicit for both sides

03

Transparent

Plain-English reasoning
+ immutable audit trail

Not just a faster dispute tool — the foundation of a trust layer between card members, merchants, and the issuer.